

Trainings ▾

Select Service Tools

Recommended Cummins® Service Tools

- No recommended Cummins® service tools

Additional Service Items

- No additional service items required

Preparatory Steps

WARNING

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Disconnect the batteries. See equipment manufacturer service information.
- Remove the alternator drive belt. Refer to Procedure 013-005 in Section 13.
(/qs3/pubsys2/xml/en/procedures/377/377-013-005.html)
- Remove the alternator. Refer to Procedure 013-001 in Section 13.
(/qs3/pubsys2/xml/en/procedures/377/377-013-001.html)
- Remove the alternator spacers. Refer to Procedure 013-036 in Section 13.
(/qs3/pubsys2/xml/en/procedures/377/377-013-036.html)
- Remove the refrigerant compressor drive belt. Refer to Procedure 009-052 in Section 9.
(/qs3/pubsys2/xml/en/procedures/377/377-009-052.html)
- Remove the refrigerant compressor. Refer to Procedure 009-051 in Section 9.
(/qs3/pubsys2/xml/en/procedures/377/377-009-051.html)
- Remove the refrigerant compressor mounting bracket. Refer to Procedure 009-055 in Section 9. (/qs3/pubsys2/xml/en/procedures/377/377-009-055.html)
- Remove the refrigerant compressor belt tensioner. Refer to Procedure 009-053 in Section 9.
(/qs3/pubsys2/xml/en/procedures/377/377-009-053.html)
- Remove the refrigerant idler pulley. Refer to Procedure 009-054 in Section 9.
(/qs3/pubsys2/xml/en/procedures/377/377-009-054.html)

- Remove the water pump belt tensioner. Refer to Procedure 008-080 in Section 8. (</qs3/pubsys2/xml/en/procedures/377/377-008-080.html>)
- Disconnect the air compressor inlet tubes. Refer to Procedure 012-109 in Section 12 (</qs3/pubsys2/xml/en/procedures/175/175-012-109-tr.html>).

Remove

Disconnect the wire harness jumper from the EGR valve motor, EGR valve position sensor, and the main engine wiring harness.

Remove the mounting capscrews from the alternator/refrigerant compressor mounting bracket.

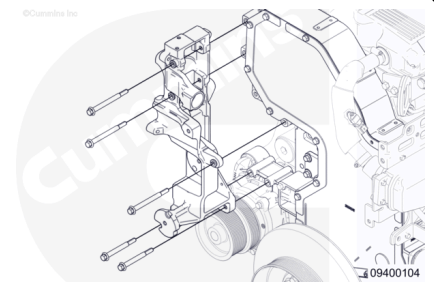
Remove the mounting capscrews from the alternator mounting bracket.

Gently pry out the bracket from the gear housing.

The split washers will create a slight friction fit between the bracket and the gear housing.

Remove the wiring harness jumper from the alternator/refrigerant compressor mounting bracket.

Remove the wiring harness jumper from the alternator mounting bracket.



Inspect for Reuse

Inspect the alternator/refrigerant compressor mounting bracket for cracks.

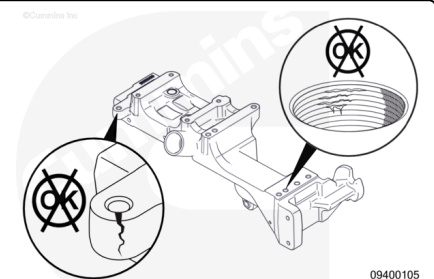
Replace the bracket, if necessary.

Inspect underneath the refrigerant compressor, alternator spacer, water pump belt tensioner, and refrigerant compressor belt tensioner for wear.

If wear is found, replace the alternator bracket.

Inspect the alternator mounting bracket for cracks.

Replace the bracket, if necessary.



Inspect underneath the alternator spacer and water pump belt tensioner for wear.

If wear is found, replace the alternator bracket.

Install

Install the wiring harness jumper to the alternator/refrigerant compressor mounting bracket.

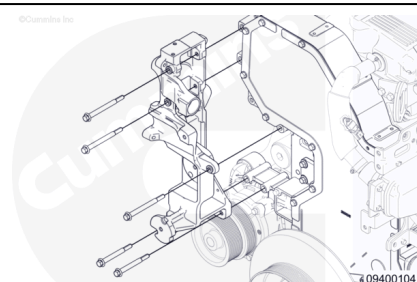
Install the wiring harness jumper to the alternator mounting bracket.

Torque Value: 18 n•m [159 in-lb]

Tap the split washers out slightly. This will ease the installation.

Install the alternator/refrigerant compressor mounting bracket to the gear housing.

Install the alternator mounting bracket to the gear housing.



Install the mounting capscrews.

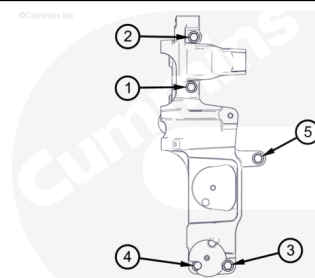
Tighten the capscrews in the sequence shown.

Torque Value:

(1, 2, 3, and 4) 36 n•m [27 ft-lb]

Torque Value:

(5) 56 n•m [41 ft-lb]



Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Connect the air compressor inlet tubes. Refer to Procedure 012-109 in Section 12. ([/qs3/pubsys2/xml/en/procedures/175/175-012-109-tr.html](http://qs3/pubsys2/xml/en/procedures/175/175-012-109-tr.html))

- Install the water pump belt tensioner. Refer to Procedure 008-080 in Section 8.
(/qs3/pubsys2/xml/en/procedures/377/377-008-080.html)
- Install the refrigerant compressor idler pulley. Refer to Procedure 009-054 in Section 9.
(/qs3/pubsys2/xml/en/procedures/377/377-009-054.html)
- Install the refrigerant compressor belt tensioner Refer to Procedure 009-053 in Section 9.
(/qs3/pubsys2/xml/en/procedures/377/377-009-053.html)
- Install the refrigerant compressor mounting bracket. Refer to Procedure 009-055 in Section 9.
(/qs3/pubsys2/xml/en/procedures/377/377-009-055.html)
- Install the refrigerant compressor. Refer to Procedure 009-051 in Section 9.
(/qs3/pubsys2/xml/en/procedures/377/377-009-051.html)
- Install the refrigerant compressor drive belt. Refer to Procedure 009-052 in Section 9.
(/qs3/pubsys2/xml/en/procedures/377/377-009-052.html)
- Install the alternator spacers. Refer to Procedure 013-036 in Section 13.
(/qs3/pubsys2/xml/en/procedures/377/377-013-036.html)
- Install the alternator. Refer to Procedure 013-001 in Section 13.
(/qs3/pubsys2/xml/en/procedures/377/377-013-001.html)
- Install the alternator drive belt. Refer to Procedure 013-005 in Section 13.
(/qs3/pubsys2/xml/en/procedures/377/377-013-005.html)
- Connect the batteries. See equipment manufacturer service information.
- Operate the engine and check for proper operation.